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Impact of frame perforations on passive cooling of photovoltaic modules: CFD analysis of various patterns

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ABSTRACT

Geometry of the frames used in a photovoltaic (PV) module affects the fluid flow and heat transfer around the system. As a method of passive cooling, various perforation patterns are introduced into the aluminum frame of a PV module, and the resulting flow field, temperature distribution, and power output are investigated in the present study. Three-dimensional computational fluid dynamics (CFD) simulations are performed in the cases of forced (wind velocity of 4 m/s) and natural convection (zero wind velocity). After comparing two types of conventional frames with a frameless module, different perforation patterns are utilized and the resulting flow field and heat transfer distributions are carefully investigated for the equivalent heat flux value of 600 W/m^2. Results show that perforations are effective in case of natural convection, in which a 3.8 K temperature drop is observed in the best case. However, the average PV temperature is almost the same in forced convection. Finally, two selected designs are studied in different typical weather conditions. As a result, the frame with 10 mm circular perforations on all sides achieves a 0.62% improvement in power generation in the temperate climate and zero wind velocity compared to the panel with a conventional frame.

1. Introduction

Renewable energy systems are getting much more attention nowadays. In 2018, 13.5% of total energy supply was provided from renewable energy sources [1]. Among different renewable energy technologies, direct electricity generation using solar photovoltaic systems (PV) is very popular. Easy installation, low operating and maintenance costs, long useful life, and the ability to produce electricity at different irradiance levels are some interesting features of PV systems. However, the main deficiency is their relatively low conversion efficiency, which is around 9–17% for conventional PV modules [2]. Additionally, PV efficiency decreases as module temperature increases, at a rate of 0.2%-0.5% reduction for each 1 °C temperature rise for different cell types [3]. Therefore, thermal management and cooling of PV modules is an important research field today [4].

Passive cooling methods try to decrease PV module temperature without extra energy consumption. Module temperature depends on meteorological parameters such as solar irradiance, ambient temperature, wind speed, and wind direction [5–11] as well as other manufacturing and installation factors such as components material, tilt angle, and installation geometry [6,12]. Accordingly, a number of researchers have tried to passively cool modules by introducing changes in the design, such as making slits or drilling holes in some parts of a PV panel. Arianamehr et al. [13] experimentally showed the cooling effect of transverse square grooves over the leading edge of a heated plate resembling a PV panel due to the change in turbulence intensity. Also, Abd-Elhady et al. [14] observed 8 °C of cooling in the numerical simulation of a naturally ventilated mono-crystalline PV by creating 9 holes of 1 mm diameter between cells. In these approaches, the objective is to increase heat transfer capacity between PV cells and ambient.

Frame is a part of module that can play an important role in heat transfer from module to the surrounding air. It can alter the air flow field around the system, and also increase the heat transfer surface area. However, these effects are usually ignored in many theoretical and numerical studies. One of the earliest studies that included frame in a numerical model is provided by Lee and Tay [15] in 2012. They used a finite element solution to find temperature distribution in PV module without modeling the surrounding air flow. They found that cells located near the frame are 5 °C cooler than the cells in the center of the module.

In 2016, Nižetić et al. [16] focused on temperature and flow field around PV modules in the presence of frame. Their experimental and

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Table 1
Specifications of the PV panel SL-50AA36 at standard test condition [16].

Number of cells in the module	36
Maximum power output	50 W ± 3%
Maximal power voltage/Open circuit voltage	17.2 V/22.9 V
Maximal power circuit current/Short circuit current	2.91 A/3.18 A
Surface area	0.55 × 0.65 m^2

Table 2
Thermo-physical properties air and PV module components.

Material	Thermal conductivity (W/m.K)	Heat capacity (J/kg.K)	Density (kg/m^3)	Layer thickness (mm)
Aluminum [19]	202.4	871	2719	–
Air [20]	0.0242	1006.43	1.225	–
Glass [9]	0.8	800	2530	3
PV cell [9]	150	700	1000	0.3
Back sheet [9]	0.35	800	800	0.3
Equivalent layer	13.196	731.66	2258.34	3.6

numerical results showed that the backside convective heat transfer is affected by a flow separation which results in a higher PV temperature. They concluded that redesigning of PV panel may reduce the observed flow separation. This research is expanded in 2018 by considering various ambient conditions [9]. According to the numerical results, they proposed that introducing some changes in the aluminum frame architecture (e.g., perforations) could result in lower PV temperature and higher efficiency. However, they did not discuss the possible negative effect of the reduced heat transfer surface area.

Recently, Nižetić et al. [17] investigated three passive cooling techniques by changing the design of a PV module. They introduced a specific perforation pattern in two sides of a PV frame and discussed the results for the wind velocity of 2.0 m/s in different directions. They concluded that this kind of perforation in the frame does not significantly affect the PV panel temperature even with different hole widths. However, a more promising cooling effect is found by introducing slits in the PV plane itself, as discussed in more detail by Kragić et al. [18].

The above literature review shows that the effect of frame geometry on PV temperature is not profoundly understood yet. The only study by Nižetić et al. [17] shows that introducing perforations into the frame can alter the temperature and flow field around a PV panel to some extent. But this phenomenon is not investigated for various perforation geometries, ambient conditions, and wind speeds (including natural convective cooling). Perforations not only affect the flow field, but also they reduce heat transfer surface area, which is a negative side effect, and depends on geometry of the holes. Accordingly, this is a complicated problem and should be tackled in more detail.

Therefore, the objective of the present study is to numerically explore the effect of frame geometry and perforations on heat transfer characteristics of PV modules. At first, a frameless PV is compared to panels with two conventional frame types, for better understanding of frame impact on heat transfer and flow field. Then, various perforation designs are modeled, and the resulting flow and temperature fields are scrutinized. Both cases of forced and natural convective cooling are considered, as the impact of perforations could be totally different. In summary, the features and novelties of the present research include:

- A three-dimensional CFD model is developed to investigate the flow field and heat transfer around a PV panel with different frame designs.
- Thermal and aerodynamic impacts of frame geometry are discussed in detail.
- Various types of perforations are introduced into the PV frame.
- Effects of perforations are studied at different ambient conditions for both natural and forced convective cooling.

2. Mathematical modeling

2.1. Problem definition

A three dimensional PV module with aluminum frame is considered to be mounted freely 0.5 m above the ground with a tilt angle of 20°. The electrical and geometrical specifications of the considered PV module (model SL-50AA36 [16]) are presented in Table 1. Different layers of the PV module (glass, PV cells, and back sheet) are modeled as a single equivalent layer with averaged thermo-physical properties. Table 2 provides the properties of air and module materials, including each PV layer, along with the resulting equivalent layer.

The computational domains shown in Fig. 1 are utilized to calculate the air flow field and temperature distributions in both PV and the surrounding air for forced and free convection cases. In forced convection cases, the wind velocity is assumed to be in the X-direction. Accordingly, the solution would be symmetric in Z-direction, and only half of the domain is applied to reduce the computational cost. All PV sides are considered as walls, with no-slip boundary conditions that are thermally coupled with the surrounding air in convection simulations. Fig. 1-a illustrates the dimensions of the computational domain in forced convection. A uniform wind velocity of 4 m/s is set to enter the domain from the inlet boundary, while a pressure outlet boundary condition is utilized at the exit. The ground surface is also considered to be a no-slip wall with the same temperature as the surrounding air, while the other sides of the domain are regarded as symmetry.

Fig. 1-b shows the computational domain in natural convection, where the wind velocity is assumed to be zero. The pressure inlet and outlet boundary conditions are considered for the lateral and the top surfaces, respectively. The ground level is also modeled as a no-slip wall at the ambient temperature.

2.2. Governing equations

2.2.1. Forced convection:

In forced convection simulations, the air flow is turbulent and Reynolds averaged Navier-Stokes (RANS) steady state equations are applied. The shear-stress transport (SST) k-ω turbulence model is chosen as it can accurately predict separation in adverse pressure gradient and has been previously approved in simulations of PV module problems [7,9,16,21]. The applied incompressible continuity and momentum equations are [22]:

$$\frac{\partial(\overline{u}_j)}{\partial x_j} = 0 \tag{1}$$

$$\overline{u}_j\frac{\partial \overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\upsilon\left[\left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i}\right) - \frac{2}{3}\frac{\partial \overline{u}_k}{\partial x_k}\delta_{ij}\right] + \frac{1}{\rho}\frac{\partial}{\partial x_j}\left[R_{ij}\right] \tag{2}$$

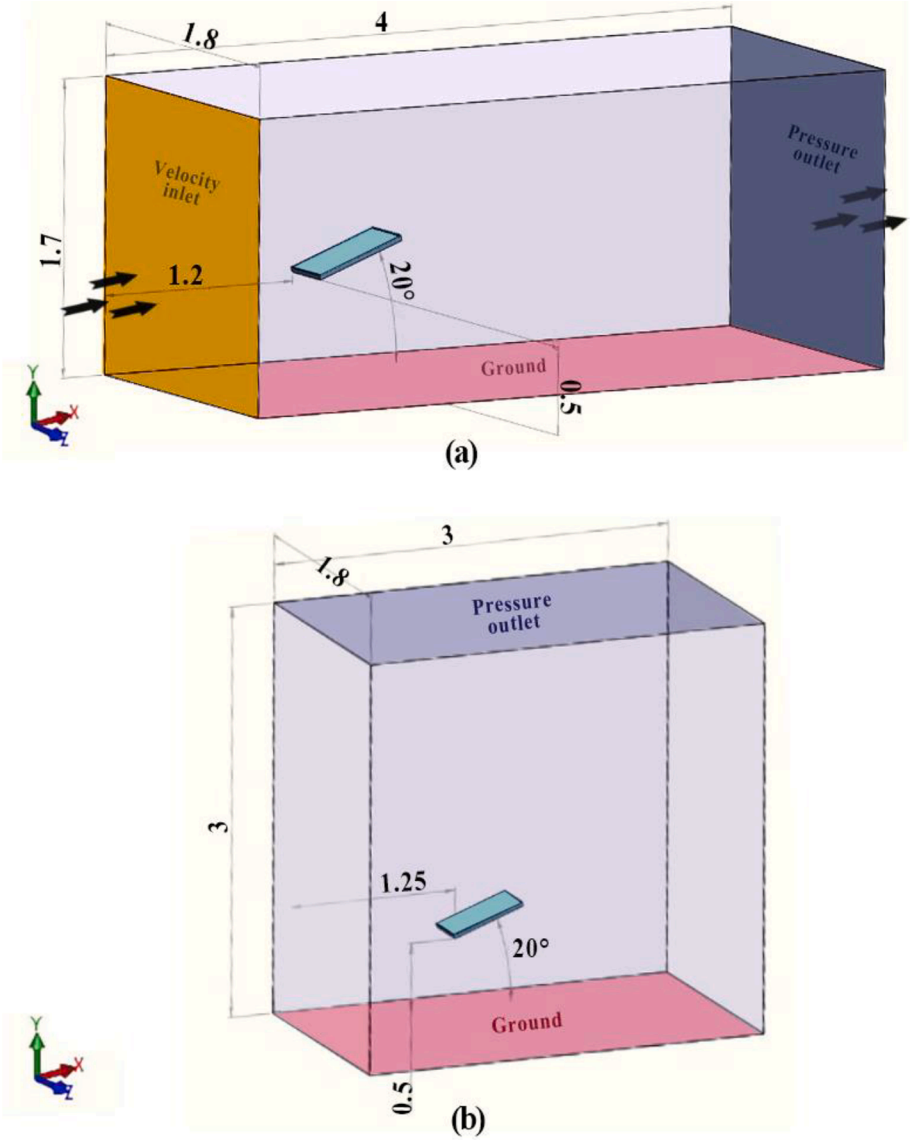


Fig. 1. The computational domain (a) forced convection (b) free convection (dimensions in m).

where ρ, $\overline{u}_i$, p and υ represent the density, velocity component, pressure, and kinematic viscosity, respectively. R_{ij} is Reynolds stress tensor according to the Boussinesq eddy viscosity assumption and is calculated as below [22]:

$$R_{ij} = -\rho\overline{u_i u_j} = \mu_t\left[\left(\frac{\partial \overline{u}_i}{\partial x_j}+\frac{\partial \overline{u}_j}{\partial x_i}\right)-\frac{2}{3}\frac{\partial \overline{u}_k}{\partial x_k}\delta_{ij}\right]-\frac{2}{3}\rho \mathrm{K}\delta_{ij} \tag{3}$$

where μ_t and K denotes turbulent viscosity and turbulence kinetic energy, which are calculated with the help of SST k-ω turbulence model.

By neglecting viscous dissipation, the energy conservation equation in air and PV is written as equations (4–5), respectively [22].

$$\overline{u}_j\frac{\partial T}{\partial x_j} = \frac{\partial}{\partial x_j}\left[(\alpha+\alpha_t)\frac{\partial T}{\partial x_j}\right] \tag{4}$$

$$\frac{\partial}{\partial x_j}\left[\frac{\partial T}{\partial x_j}\right]+\frac{Q}{k}=0 \tag{5}$$

where α, α_t, T and Q indicate thermal diffusivity, eddy thermal diffusivity, temperature, and the volumetric heat source. This source term is applied to model the absorbed solar irradiance, which is dissipated to ambient by means of convection. As a result, there would be no need to model complicated radiation models such as the discrete ordinates (DO) method. However, the appropriate value for this term should be carefully calculated by reducing the radiative heat losses and also the electrical output power from the total absorbed solar energy flux. This is done by writing an integral energy balance equation for the PV module in steady state condition:

$$\alpha G_t = \frac{P_{out}}{A}+q''_{rad-sky}+q''_{rad-ground}+q''_{eq} \tag{6}$$

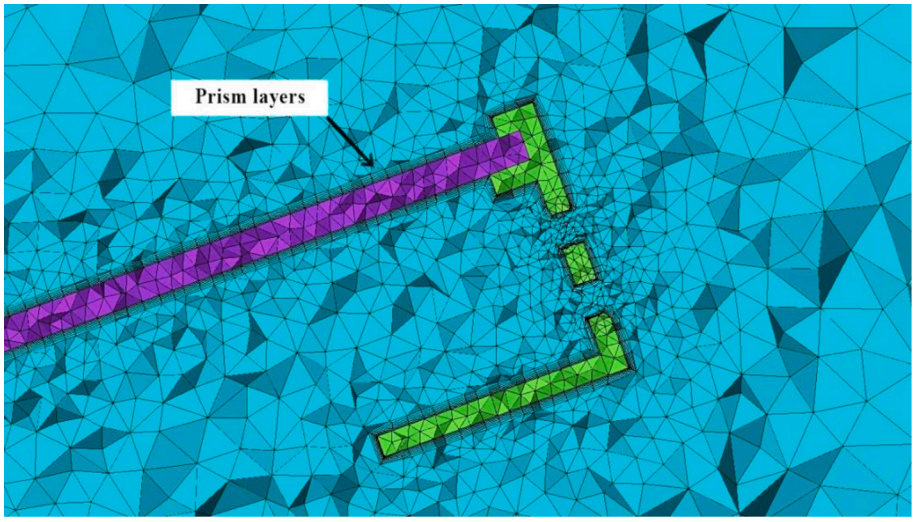


Fig. 2. A sample of the computational grid cross-section near frame perforation.

where α and G_t are absorption coefficient and total solar irradiance on the tilted PV surface. q''_{eq} is the equivalent heat flux that is dissipated through the PV surfaces and frame by means of convection heat transfer. The electrical power (P_{out}) in the above equation is determined by [6]:

$$P_{out} = \eta_{ref}\left[1-\beta_{ref}\left(T_{pv}-T_{ref}\right)\right]AG_t \tag{7}$$

where η_{ref} and β_{ref} are electrical efficiency and temperature coefficient at the reference temperature ($T_{ref} = 25^{\circ}$C), with the corresponding values of 0.154 and 0.0047 $^{\circ}C^{-1}$, respectively. A is the module surface area exposed to solar irradiance. Radiative losses into sky and ground are given by:

$$q''_{rad-sky} = \varepsilon\sigma F_1\left(T_{pv}^4 - T_{sky}^4\right) \tag{8}$$

$$q''_{rad-ground} = \varepsilon\sigma F_2\left(T_{pv}^4 - T_{ground}^4\right) \tag{9}$$

In the above equations, σ is Stephan-Boltzmann constant, F_1 and F_2 are shape factors for the front and back surface of PV and ε is emissivity, which is considered to be 0.85 in the present study. The sky temperature T_{sky} is also obtained by [23]:

$$T_{sky} = 0.0552 \times T_{amb}^{1.5} \tag{10}$$

Accordingly, the equivalent heat flux (q''_{eq}) can be found by solving equations (6–10) to calculate the volumetric heat source (Q) in the energy equation. Kragić et al. [9] showed that this equivalent heat flux method can predict temperature distributions that are in good agreement with experimental measurements. In the present study, q''_{eq} is first considered to be 600 W/m^2 for all cases with the ambient temperature of 300 K for comparison. Later the appropriate value is applied in three typical weather conditions as described in section 4.4.

2.2.2. Natural convection:

In the absence of wind, the free convective flow is determined to be laminar in the present case. As temperature variations are not large, the Boussinesq approximation is implemented for the buoyancy force. Accordingly, the continuity, momentum, and energy equations in the surrounding air can be written as [24]:

$$\frac{\partial u_j}{\partial x_j} = 0 \tag{11}$$

$$u_j\frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho_0}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\nu\frac{\partial u_i}{\partial x_j}\right) + g\beta(T-T_0)\delta_{i2} \tag{12}$$

$$u_j\frac{\partial T}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\alpha\frac{\partial T}{\partial x_j}\right) \tag{13}$$

where α and β are thermal diffusivity and thermal expansion coefficients. The Kronecker delta function δ_{i2} implies that the buoyancy force is only included in the Y-direction perpendicular to the ground.

2.3. Numerical procedure

The above mentioned governing equations along with the boundary conditions described in section 2.1 are solved with finite volume method using Ansys Fluent 19. The second-order upwind scheme is applied to discretize pressure, momentum and, energy equations, and SIMPLE (Semi- Implicit Method for Pressure Linked Equations) algorithm is used to solve velocity–pressure coupling equations. To prevent the solution procedure from divergence, typical under relaxation factors of 0.3, 0.7, and 1 are utilized for pressure, momentum, and energy equations, respectively. Also, convergence criterion for the continuity, momentum, and energy equations are considered to be 10^{-5}, 10^{-4} and 10^{-8}, respectively. Further continuation of iterations does not change the calculated results, and a well converged solution could be obtained.

An unstructured grid with a very fine stretch near the solid boundaries is employed, as illustrated in Fig. 2. Tetrahedral elements are applied in the domain, while prism layers are used in the boundary layer adjacent to PV borders. The prism layers include 7 to 10 cells perpendicular to the surface with an initial height of 0.1 mm and a growth rate of 1.15. The number of computational cells in the holes and grooves of the frame should be quite enough to have a reliable solution.

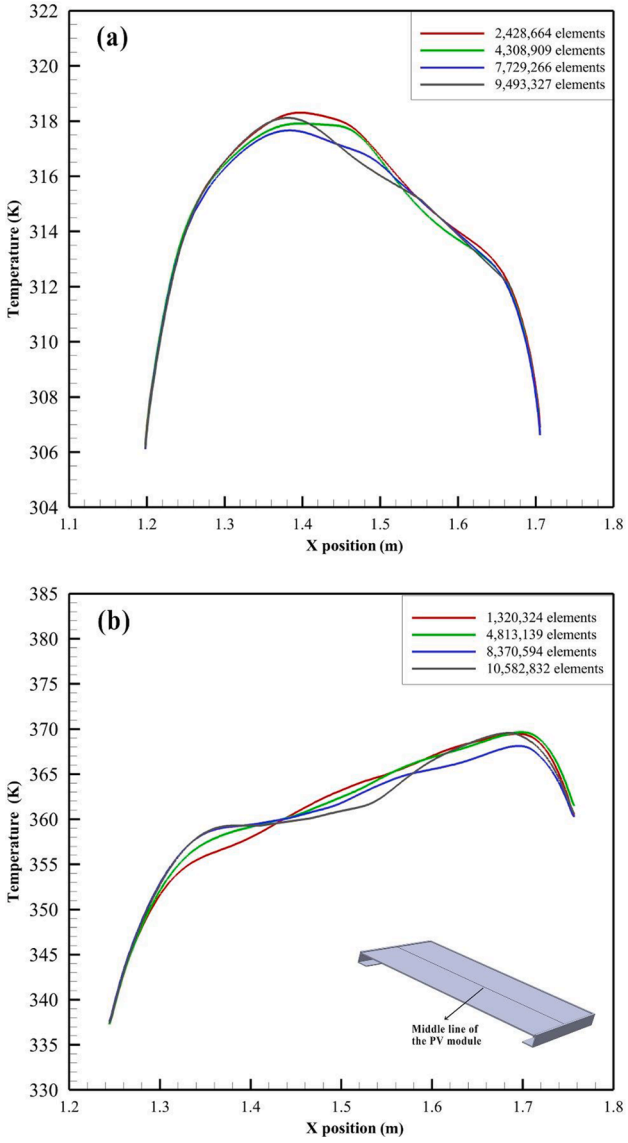


Fig. 3. Grid independency test on the middle line of the module with frame 1 (q''_{eq} = 600 W/m^2 and T_{amb} = 300 K) (a) forced convection (b) natural convection.

Table 3
Grid independency test for the module with frame 1 in forced and natural convection (q''_{eq} = 600 W/m^2, T_{amb} = 300 K).

Force convection (V = 4 m/s)		Natural convection (V = 0 m/s)	
Number of cells	Average PV temperature (K)	Number of cells	Average PV temperature (K)
2,428,664	314.5	1,320,324	358.28
4,308,909	314.4	4,813,138	358.64
7,729,266	314.3	8,370,594	358.60
9,493,327	314.3	10,582,832	358.60

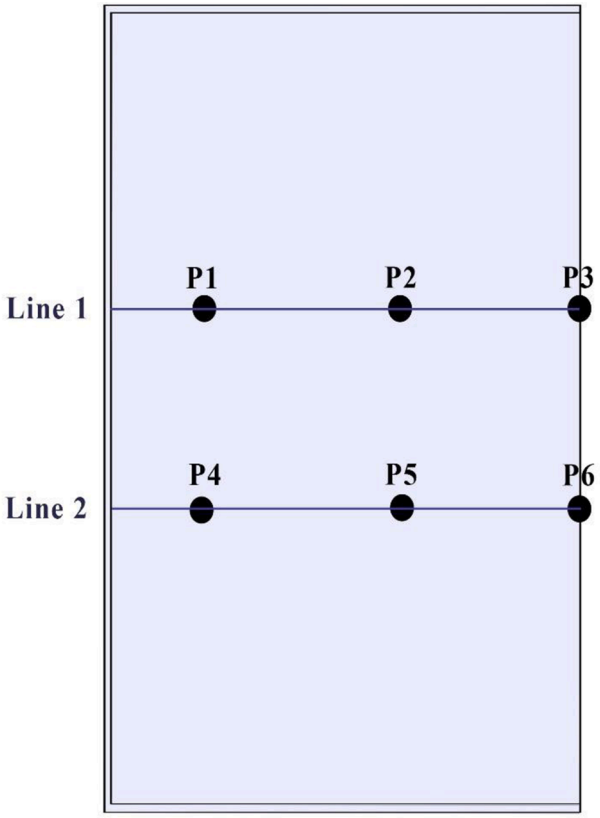


Fig. 4. Monitoring lines in the present study as well as the observed points of Kragić et al. [9].

Accordingly, grid studies are performed to ensure that the calculated results are independent of cell numbers. For example, Fig. 3 compares the calculated temperature distributions on the middle line of PV with frame 1 (defined in section 3) in X-direction for different numbers of computational cells at q''_{eq} = 600 W/m^2 and T_{amb} = 300 K in forced and free convection cases. The corresponding average PV temperatures are also presented in Table 3. As the resulting PV temperatures are almost the same for the two finest computational grids, the selected number of cells in forced and natural convection are 7,729,266 and 83,370,594, respectively in this case.

After obtaining the converged solution, the distribution of heat flux is acquired to calculate the averaged convective heat transfer coefficient at each surface defined by:

$$h_{conv} = \frac{Q_s}{A(T_s - T_{amb})} \tag{14}$$

In the above equation, Q_s is the total convective heat transfer from any desired surface and T_s is the corresponding averaged surface temperature. The resulting h_{conv} is going to be used as an indicator to investigate the effects of perforations introduced in the frame geometry.

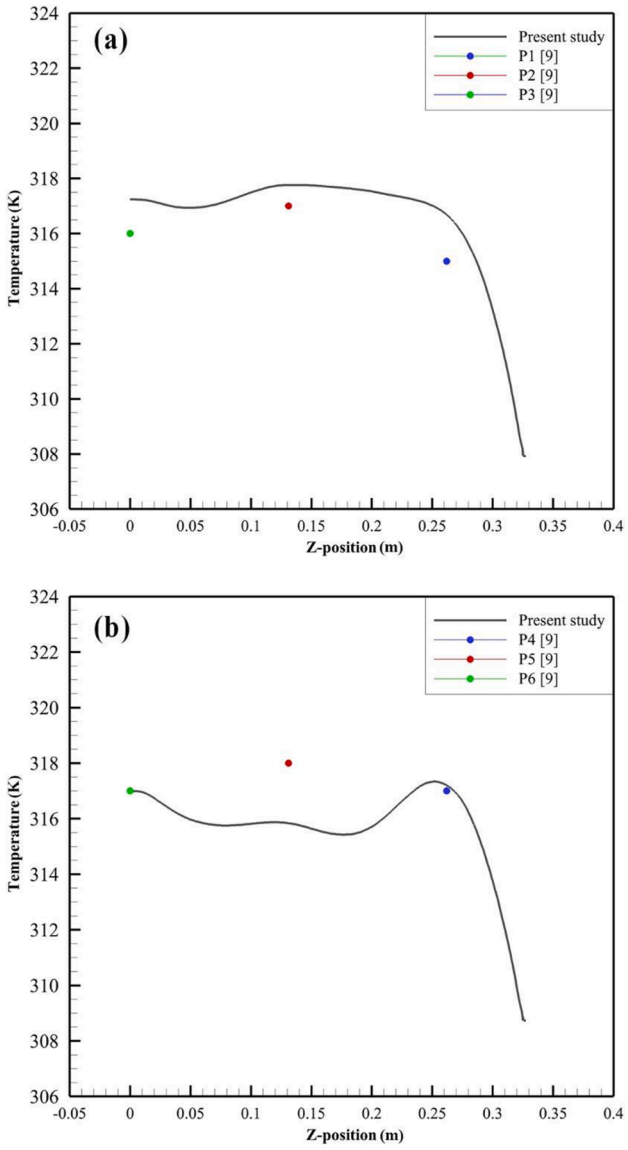


Fig. 5. Temperature distribution on the backboard surface of PV on (a) Line 1 (b) Line 2 in the present study and comparison with results of Kragić et al. [9].

Table 4
Comparison of the resulting average PV temperature with two implicit correlations proposed by Coskun et al. [25] (q''_{eq} = 600 W/m^2, V = 4 m/s, T_{amb} = 300 K).

Method	PV temperature (K)
Present study	314.3
Modified Akyuz et al. correlation [25]	314.1
Modified Chenni et al. correlation [25]	313.5

2.4. Validation

To validate the accuracy of the present calculations, the PV module considered by Kragić et al. [9] is simulated in forced convection (V = 4 m/s) with the equivalent heat flux of q''_{eq} = 600 W/m^2 in the ambient temperature of 300 K. At first, the present results are compared with the numerical simulation of Kragić et al. [9] at different monitoring points located at the backboard sheet as illustrated in Fig. 4. Fig. 5 shows that the present temperature distributions on lines 1 and 2 agree well with the data for points 1–6 provided by Kragić et al. [9]. Although the equivalent single layer model is utilized in the present simulation, the average temperature difference in the six points is 0.3 °C with the maximum deviation of 2.1 °C observed at P5.

The average PV temperature, in this case, is also compared with two experimentally modified implicit correlations proposed by Coskun et al. [25] in 2017. The corresponding equations known as modified Akyuz et al. and modified Chenni et al. are given by [25]:

$$T_{PV} = 0.95.T_{amb} + 0.0674 + 0.0214608.G_t - 0.0000041286.G_t^2 - 0.3.V \quad (15)$$

$$T_{PV} = T_{amb} - 1.93666 + 0.0138.G_t.(1 + 0.031.T_{amb})(1 - 0.042.V) + 0.007882.G_t - 0.0000134647.G_t^2 \quad (16)$$

where T_{amb} and T_{PV} indicate ambient and module temperatures in °C. V is wind speed and G_t is the global incident solar irradiance in W/m^2. Table 4 compares the temperature predicted by these two correlations with the present numerical result. An excellent agreement is observed with the maximum deviation of 0.8 °C perceived in the modified Chenni et al. correlation.

3. Frame geometry and perforation patterns:

The cross-sectional area and dimensions for the two types of conventional frames are given in Fig. 6. Frame 2 is usually employed in modules with large dimensions and weight. The three different sides of the frames are identified in Fig. 7 with the dimensions of L = 0.55 and W = 0.325 m.

The geometric specifications for five different patterns (a-e) investigated in this study are introduced in Fig. 8. By applying these patterns to the three different sides of frame 1, a total number of 13 frame designs are studied in this research. These designs are divided into three categories according to the number of perforated sides as provided in Table 5.

4. Result and discussion

4.1. Conventional frames with no perforations

PV frame affects the convective heat transfer to the ambient in two ways: It alters the flow field around the PV plane specifically at the backside, and on the other hand, it increases the heat transfer surface area similar to fins. To better understand this phenomenon, the PV module is simulated with the two different conventional frames defined in Fig. 6, and results are compared with a similar frameless PV in both forced and free convection cases.

Fig. 9 compares the velocity distribution in the symmetry plane of the three modules in case of forced convection with wind velocity of V = 4 m/s, equivalent heat flux of q''_{eq} = 600 W/m^2, and the ambient temperature of T_{amb} = 300 K. Clearly, the separated wake zone at the backside of the two modules with frames is larger than the wake behind the frameless PV. Accordingly, the reduction of convection heat transfer coefficient (h) at the backside of the modules with frames 1 and 2 is expected. Table 6 compares the h values for the three cases along with the corresponding convective heat transfer from the back sheet to the ambient. These results show that an approximately 30% reduction has occurred in the heat transfer coefficient on the backside due to the presence of frames. However, the temperature distribution is also affected by the fin-like impact of frames and should be investigated.

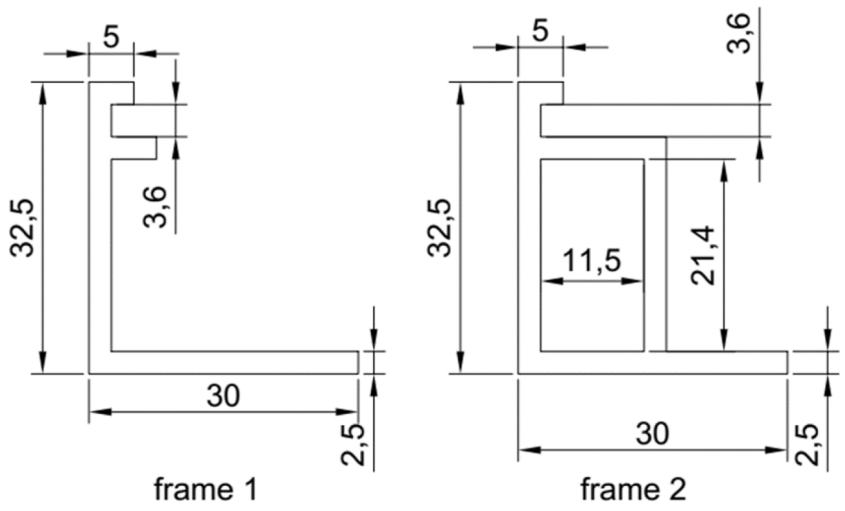


Fig. 6. Cross-sectional area and dimensions (mm) of conventional frames.

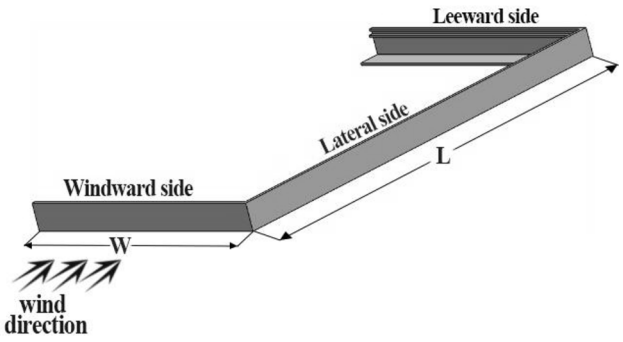


Fig. 7. The three different sides of a frame.

Fig. 10 illustrates the temperature distribution on the backside surface of the three modules under investigation. Although the cooler zones near the perimeter of the framed modules are evident due to the fin-like performance of the frames, the hot zones on the frameless PV are generally smaller because of better backside heat transfer coefficient. Table 6 reveals that these two effects almost cancel each other, as the average PV temperature is approximately identical for the three modules. The smaller backside convective heat dissipation (Q_{back}) in the framed modules is compensated by the heat transfer from frames 1 and 2. To be more specific, frame 2 not only has a larger surface area, but also its contact area with PV perimeter is more, which results in better heat dissipation through the frame. However, the resulting average PV temperature is only 0.3 K cooler. These observations promote the idea that a change in the frame design can be helpful if it can enhance the backside heat transfer coefficient without much affecting the heat dissipation through the frame surface.

In case of natural convection (zero wind velocity), the three modules are again simulated with similar conditions, (q''_{eq} = 600 W/m^2, T_{amb} = 300 K, tilt angle $\beta = 20^\circ$) for the sake of comparison. Fig. 11 indicates that the fin-like performance of frames 1 and 2 on the perimeter of a naturally cooled module is very effective in decreasing the average PV temperature. Table 7 reports that although a 21% reduction occurs on the backside heat transfer coefficient (h) in the framed modules, they are about 3.5 K cooler due to the enhanced heat dissipation through frames. Therefore, the presence of frames is favorable in these cases of natural convection. Accordingly, a change in the frame geometry would be desirable if the probable increase in the backside h is more effective than the reduction in heat transfer through the frame. This is going to be discussed in the next subsection.

4.2. Perforated frames in forced convection

Now, the different frame designs introduced in Table 5 are employed and simulations are performed in the case of forced convection. The resulting impact on flow field, backside heat transfer coefficient, heat transfer from frame, and consequently the PV temperature distribution are investigated carefully and reported in Table 8. For better understanding of these effects, the results are categorized into different subsections, according to the number of perforated sides.

4.2.1. Perforations in all three sides

Fig. 12 shows the temperature distribution in the PV modules with frames 3–7 in which patterns a-e are introduced in all of the three sides. At the first glance, a hot zone in the PV with frame 7 is noticeable, while the temperature distributions in the other modules are almost similar. Results of frames 3–7 in Table 8 reveal that none of these modifications could achieve a colder average PV temperature. Although the heat transfer coefficient on the frame surface increases in perforated geometries, the backside convective coefficient is marginally reduced. As a result, the heat dissipation through frame surfaces is increased in frames 3–6 even though the surface area is reduced, while the overall PV experiences a very small temperature increase in comparison with the conventional frame. Besides, frame 7 shows a poor performance due to the large reduction in surface area, although it achieves the highest heat transfer coefficient among these cases.

A careful examination of the flow field is necessary to see that why these changes could not increase the backside heat transfer coefficient. Fig. 13 shows the velocity vectors on perforated sides of frames 3–7. The air flow can easily enter through the perforations on the windward side, but also the reversed flow in the separated zone enters the backside area from the leeward perforations. Conceivably, these two counter flow streams may reduce the flow velocity near the backside surface and

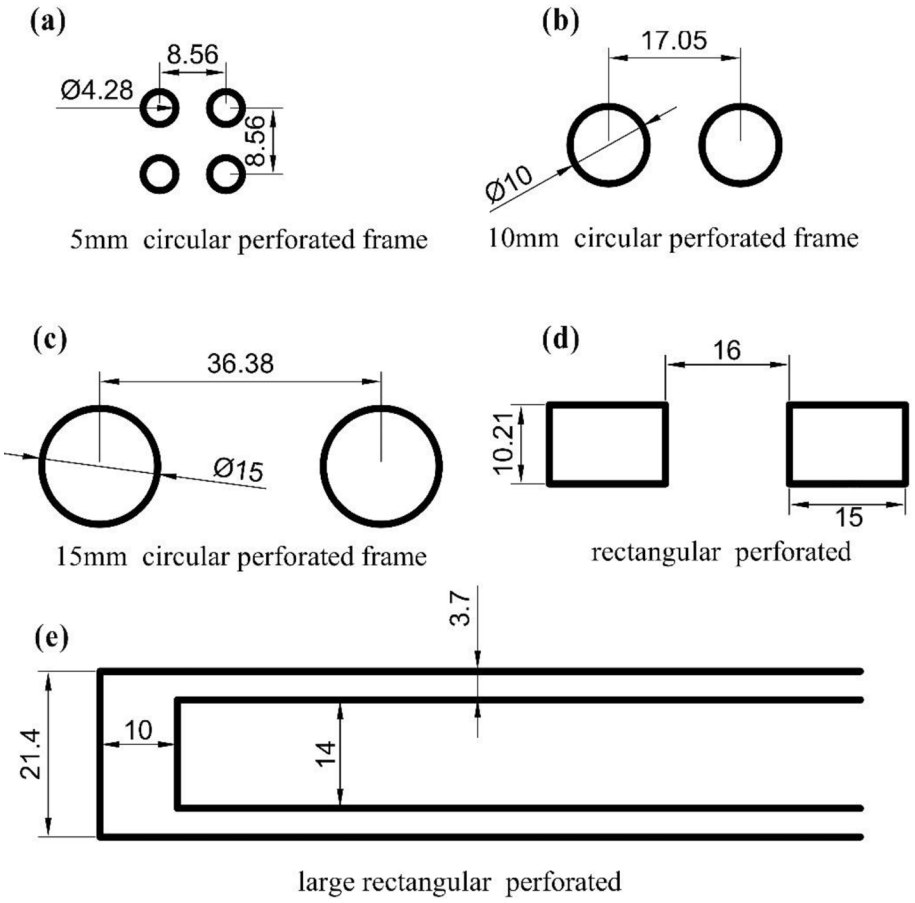


Fig. 8. Geometric specifications of perforation patterns a-e (dimensions in mm).

curtail the convective heat transfer coefficient.

In response to this observation, only half-length of the leeward side is perforated in frames 8–9 with similar patterns corresponding to frames 7 and 4, respectively. Table 8 shows that the heat transfer coefficient at the backside surface improves in both new cases, while the heat transfer coefficient on the frame side is reduced. The total frame surface area is also larger in frames 8–9. Therefore, both cases could achieve a 0.4 K temperature drop in comparison to frames 7 and 4. The corresponding temperature distributions are also illustrated in Fig. 14. As a result, the average PV temperature in the module with frame 8 is 0.2 K colder than the conventional frame 1.

4.2.2. *Perforations in two sides*

Another option is to keep one side intact and incorporate perforations only on two sides. The windward and lateral sides of frames 10 and 11 are perforated, and temperature distributions are compared with their associated full perforated frames of 7 and 4 in Fig. 15. Table 8 reveals that the backside heat transfer coefficient is not as high as the half-length perforated frames of 8 and 9. However, the extra frame surface area resulted in a slight temperature drop.

The other possible design is to perforate windward and leeward sides in frames 12 and 13. Fig. 16 demonstrates how the velocity vectors in the separated zone can enter the backside region from the leeward side holes. Accordingly, the lowest backside heat transfer coefficient is observed among the considered cases as reported in Table 8. The corresponding temperature distributions are compared in Fig. 17, which shows the higher temperatures in this type of perforations. These results are consistent with the findings of Nižetić et al. [17] in which this type of perforations was not found to be satisfactory.

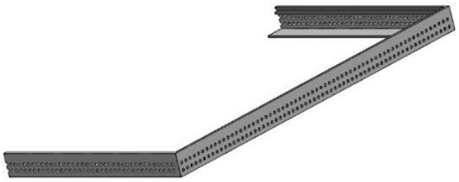
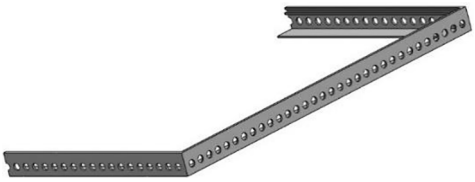
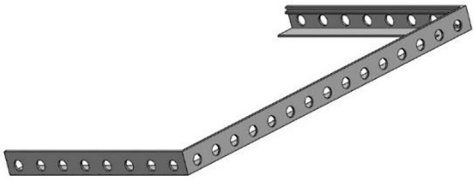
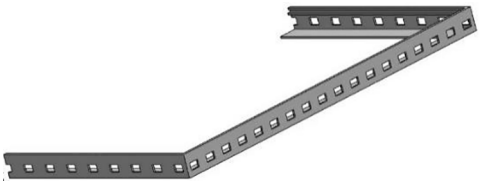
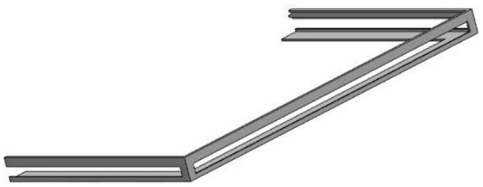
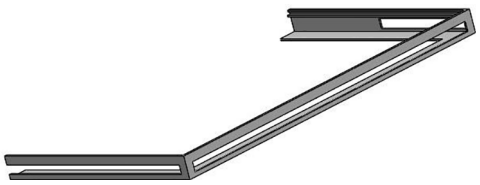
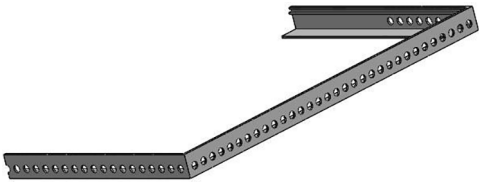
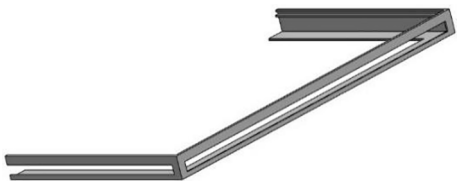
4.2.3. *Perforations in one side*

In frames 14 and 15, perforations are only introduced into the lateral side. Table 8 shows that the backside heat transfer coefficient is better than full three side perforations. In addition, the frame surface area is larger, which would result in overall better passive cooling. To be more specific, the PV module with frames 14 and 15 are 1.4 K and 0.4 K cooler than their corresponding three side perforated frames 7 and 5. The temperature distributions in this comparison are provided in Fig. 18.

4.2.4. *Summary of forced convection results*

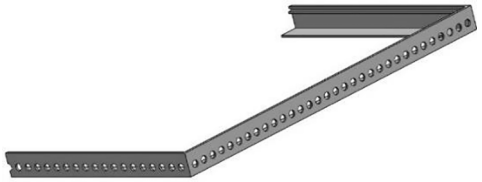
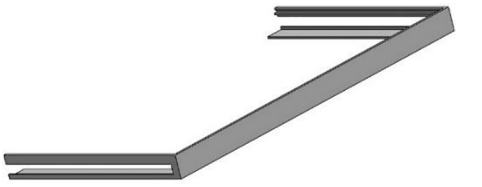
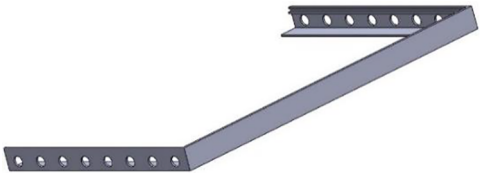
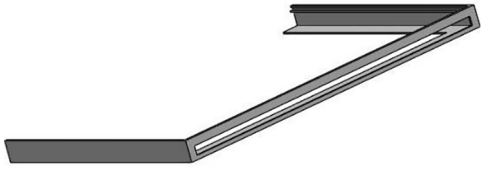
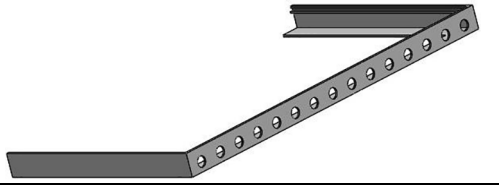
Fig. 19 compares the resulting average PV temperature for all of the considered frame geometries in forced convection. Only four frame geometries could achieve a marginally lower temperature than the conventional frame, with the lowest record corresponding to frame 9, which is just 0.2 K colder than frame 1. The reason is that the reversed flow from the leeward side perforations suppressed the velocity field in the backside region in most cases. Moreover, the frame heat transfer area is reduced due to perforations. Therefore, it can be concluded that frame perforations do not have a considerable impact on forced convective cooling. However, the thermal performance can be totally different in natural convection, which is going to be discussed in the next section.

Table 5
Definition of different proposed frame designs.

Design name	Pattern	Perforated sides			shape
		Windward side	Lateral side	Leeward side	
Frames with three perforated sides					
Frame 3	(a)	✓	✓	✓	
Frame 4	(b)	✓	✓	✓	
Frame 5	(c)	✓	✓	✓	
Frame 6	(d)	✓	✓	✓	
Frame 7	(e)	✓	✓	✓	
Frame 8	(e)	✓	✓	✓	
Frame 9	(b)	✓	✓	✓	
Frames with two perforated sides					
Frame 10	(e)	✓	✓	×	
Frame 11	(b)	✓	✓	×	

(*continued on next page*)

Table 5 (*continued*)

Design name	Pattern	Perforated sides			shape
		Windward side	Lateral side	Leeward side	
Frames with three perforated sides					
Frame 12	(e)	✓	×	✓	
Frame 13	(c)	✓	×	✓	 
Frames with one perforated side					
Frame 14	(e)	×	✓	×	
Frame 15	(c)	×	✓	×	

4.3. Perforated frames in natural convection

When the ambient wind velocity is zero, the free convective cooling of the PV module is very important. The physics of fluid flow and heat transfer differs from the forced convection, and perforations have a greater impact on passive cooling. Therefore, all of the proposed frames are now simulated in case of natural cooling, with the same value of q''_{eq} = 600 W/m^2 for the sake of comparison. The simulation results are presented and discussed in the following subsections according to the number of perforated sides.

4.3.1. Perforations in all three sides

Fig. 20 shows the resulting temperature distributions in natural convection cooling for PV modules with frames 3–7 and compares with the conventional frame 1. Clearly, perforations could achieve lower temperature distributions, except frame 7 in which a large part of the frame area is removed. The best performance belongs to frame 4, in which hot air can easily flow through perforations without much reducing the heat transfer surface area.

According to Fig. 21-a, the hot air gets stuck near the backside surface of the module, due to the presence of the leeward side of frame 1, and a thick boundary layer with poor cooling performance is observed. When perforations are introduced, especially on the leeward side, the trapped warm air can be removed more easily, which results in higher heat transfer coefficients and lower module temperatures.

Table 9 summarizes the heat transfer performance of different frame geometries. The heat transfer coefficient on the front surface is improved in frames 3–7, while the backside heat transfer coefficient is almost unchanged, except in frame 7 with the large opening. h_{frame} is also improved significantly in these cases. As a result, the average PV temperature is cooler in frames 3–6, with a dramatic drop of 3.8 K in frame 4. The higher values of heat transfer coefficients in frame 7 are not effective, due to the small frame area in this case.

The half-length leeward side perforations in frames 8–9 are not effective in case of natural cooling (in contrast to the forced convection). Table 9 reveals that the average PV temperature with frame 9 is 1.3 K higher than the corresponding full perforated frame 4, which is due to the lower h_{frame}, although more surface area is available. On the other

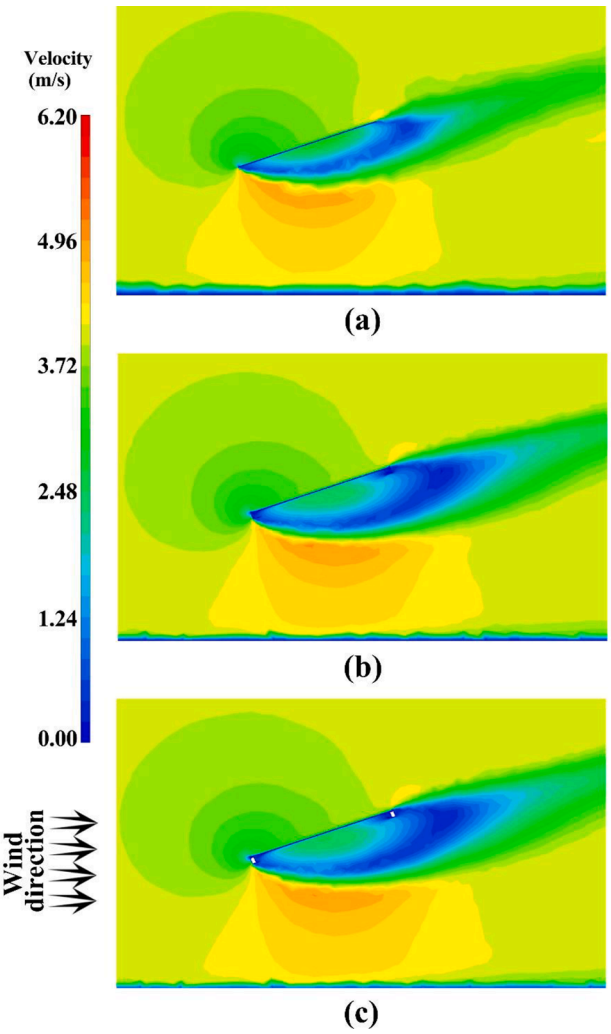


Fig. 9. Velocity distribution for (a) frameless (b) frame 1 (c) frame 2 on the symmetry plane at V = 4 m/s.

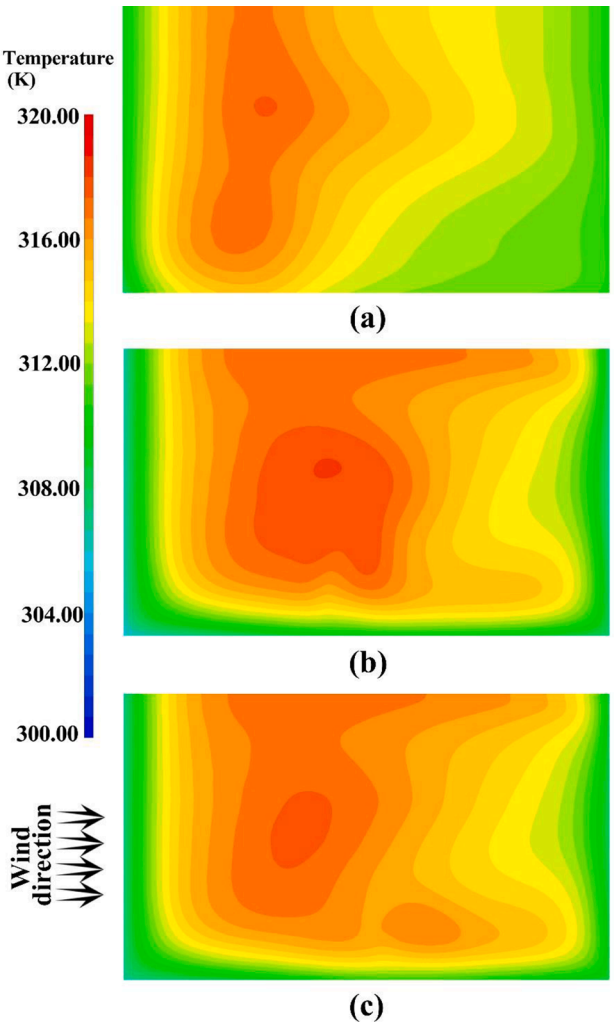


Fig. 10. Temperature distribution on back surface of PV module in (a) frameless (b) frame 1 (c) frame 2 at V = 4 m/s.

Table 6
Heat transfer results for the frameless and framed modules in forced convection, V = 4 m/s.

Design name	T_{avg} (K)	h_{back} (W/m^2.K)	Q_{back} (W)	Q_{frame} (W)	A_{frame} (cm^2)
Frameless	314.0	23.0	57.9	–	–
Frame 1	314.3	16.3	41.0	17.5	1648.5
Frame 2	314.0	16.7	39.6	20.4	2291.0

hand, the extra surface in frame 8 would cause a 37% higher heat removal through frame surface in comparison to frame 7, and the resulting average temperature in frame 8 reduces by 1.7 K accordingly.

4.3.2. Perforations in one or two sides

When the leeward side is kept solid in frames 10–11, the front surface heat transfer coefficient is reduced in comparison to the full perforated frames in natural convection. Accordingly, the thermal performance of frame 11 is not as well as frame 4, although it has a larger heat transfer surface area. On the other hand, when the lateral side is not perforated in frames 12–13, the extra heat transfer area on this side is beneficial, as it does not block the outflow of hot air in case of natural convection cooling. Therefore, frame 13 could achieve the lowest average PV temperature among the studied frames in natural convection. The corresponding temperature distributions are provided in Figs. 22-23, along with the numerical values provided in Table 9. It is interesting to note that frame 13 that achieved the best thermal performance in natural convection is not the preferred choice in forced convection due to the reversed flows through leeward perforations.

The two remaining cases are frames 14–15, in which the lateral side is only perforated. Table 9 shows that the heat transfer coefficients are not significantly improved in comparison with the conventional frame 1, and the resulting temperature drop in frame 15 is not as well as the full perforated frame 5 with the same pattern, although it has a larger heat transfer area. The corresponding temperature distributions in Fig. 24 illustrate the hotter zones near the leeward side with no perforations in frames 14 and 15.

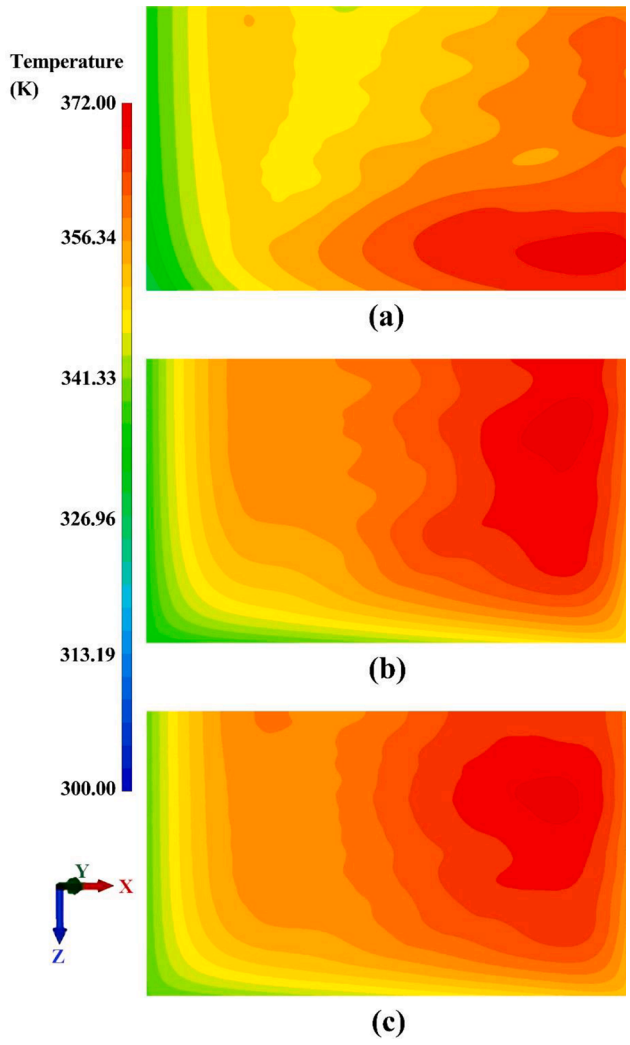


Fig. 11. Temperature distribution on back surface of PV module (a) frameless (b) frame1 (c) frame 2, at V = 0 m/s.

Table 7
Heat transfer results for the frameless and framed modules in natural convection, V = 0 m/s.

Design name	T_{ave} (K)	h_{back} (W/m^2.K)	h_{front} (W/m^2.K)	Q_{back} (W)	Q_{frame} (W)
Frameless	362.1	3.7	5.7	40.9	–
Frame 1	358.6	2.9	5.2	29.6	24.4
Frame 2	358.3	2.9	5.2	28.4	25.9

4.3.3. Summary of natural convection results

Fig. 25 shows and compares the resulting average PV temperatures in all frame designs in natural convection cooling. Frame 4 has the lowest temperature with a 3.7 K drop compared to frame 1, which is a significant amount. Almost all cases are colder than the conventional frame, except the frameless PV and those with "pattern e" perforations, which have a very low heat transfer surface area. In summary, the geometries with the leeward side perforations prevent the hot zone trapped behind the module by facilitating the upward natural flow through the holes.

Table 8
Heat transfer results of perforated frames at V = 4 m/s.

Perforated side	Design name	T_{avg} (K)	h_{back} (W/m^2.K)	Q_{frame} (W)	h_{frame} (W/m^2.K)	A_{frame} (cm^2)
Three sides	Frame 1	314.3	16.3	17.5	17.5	1648.5
	Frame 3	314.3	16.1	18.8	21.4	1657.9
	Frame 4	314.5	15.9	18.3	20.5	1591.4
	Frame 5	314.6	15.9	18.5	21.0	1573.1
	Frame 6	314.5	16.2	18.1	20.0	1585.4
	Frame 7	315.9	16.3	13.4	24.7	1383.1
	Frame 8	315.5	17.1	14.8	24.2	1428.4
	Frame 9	314.1	17.1	18.0	18.4	1600.6
Two sides	Frame 10	315.6	16.7	15.4	23.6	1455.2
	Frame 11	314.2	16.8	17.8	20.8	1607.9
	Frame 12	316.2	14.6	16.6	23.0	1504.2
	Frame 13	314.8	15.7	18.5	20.5	1609.1
One side	Frame 14	314.5	16.8	15.0	17.7	1527.3
	Frame 15	314.2	16.8	17.4	17.7	1612.6

4.4. Performance of selected frames in typical weather conditions

After careful CFD analysis of different frame designs in $q''_{eq} = 600$ W/m^2, two frames with better thermal performance are now selected to be studied in sample weather conditions. Frame 4 is selected due to having the lowest module temperature in natural convection, and frame 9 is chosen because of increasing the heat transfer coefficient at the backside surface of the PV in forced convection.

Table 10 provides three typical ambient conditions for summer (21 July), temperate (21 November), and winter (5 February) days in Shiraz, Iran, given by the hourly data in Meteonorm 7.3 software [26]. The resulting temperature and power output of the PV module with selected modified frame designs are going to be compared with the conventional frame 1. In each case, the appropriate value for q''_{eq} should be calculated according to equation (6). In case of natural convection, the higher PV temperatures increase radiative heat loss, which consequently result in lower values of q''_{eq} to be dissipated by natural convection.

Table 11 indicates that module temperatures with different frames are practically the same at the wind speed of 4 m/s and the electrical power per unit area in frame 9 has only increased by 0.11%, 0.07%, and 0.09% in winter, temperate, and summer days, respectively. On the other hand, the module with frame 4 experiences about 1 K cooler temperatures in natural convection, which results in an increase of 0.50%, 0.62%, and 0.59% in power generation, for winter, temperate, and summer days, respectively. As a result, utilization of frame 4 could be beneficial thorough out a year. It does not impose higher temperatures in windy conditions and cools the module at low wind speeds. It also reduces the module weight as well as the aluminum consumption.

5. Conclusion

Various perforation patterns are introduced into the aluminum frame of a PV module, and the resulting temperature distribution and power output are compared with conventional frame designs. Three-dimensional CFD simulations are performed in forced convection (wind velocity of 4 m/s) and natural convection (zero wind velocity). At first, a comparison is made between a module with two conventional

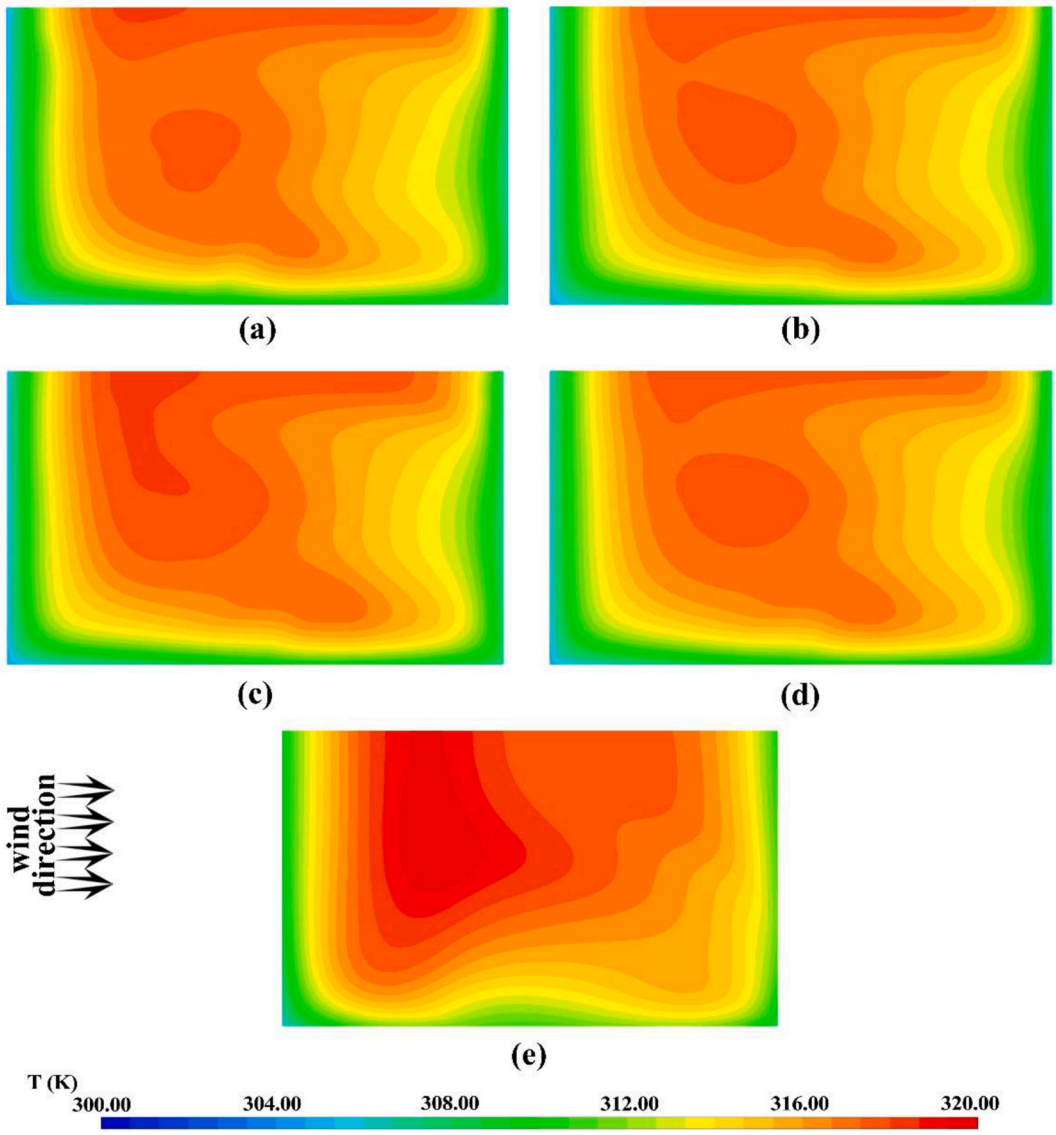


Fig. 12. Temperature distribution on back surface of PV module with (a) frame 3 (b) frame 4 (c) frame 5 (d) frame 6 (e) frame 7, at V = 4 m/s.

frame geometries and a frameless module to see how frames affect the convective heat dissipation. Then, different perforated frames are utilized, and the resulting flow field and heat transfer results are carefully investigated for a constant value of $q''_{eq} = 600$ W/m^2. Finally, two selected designs are studied in typical weather conditions for three summer, temperate, and winter days. For the specific systems investigated in the present study, it is concluded that:

- The presence of the frame in case of forced convection reduces the backside heat transfer coefficient by 29.1% and 27.4% in PV modules with frames 1 and 2, respectively.
- The extra heat transfer area of frames almost compensates for the reduction in backside heat transfer coefficient in forced convection, and the resulting average PV temperature is approximately the same as the frameless module.
- The fin-like performance of conventional frames is more effective in case of natural convection, although the backside heat transfer coefficient is 21% less than a frameless module.
- Perforations increase the heat transfer coefficient on the frame surface. However, the backside coefficient in forced convection is not improved in fully perforated cases due to the occurring reversed flow from the holes in the leeward side.
- In forced convection, the backside heat transfer coefficient can be improved slightly in frames with partial or with no perforations on the leeward side.
- On the average, no significant temperature drop is observed in most frames in forced convection. Nevertheless, perforations with “pattern

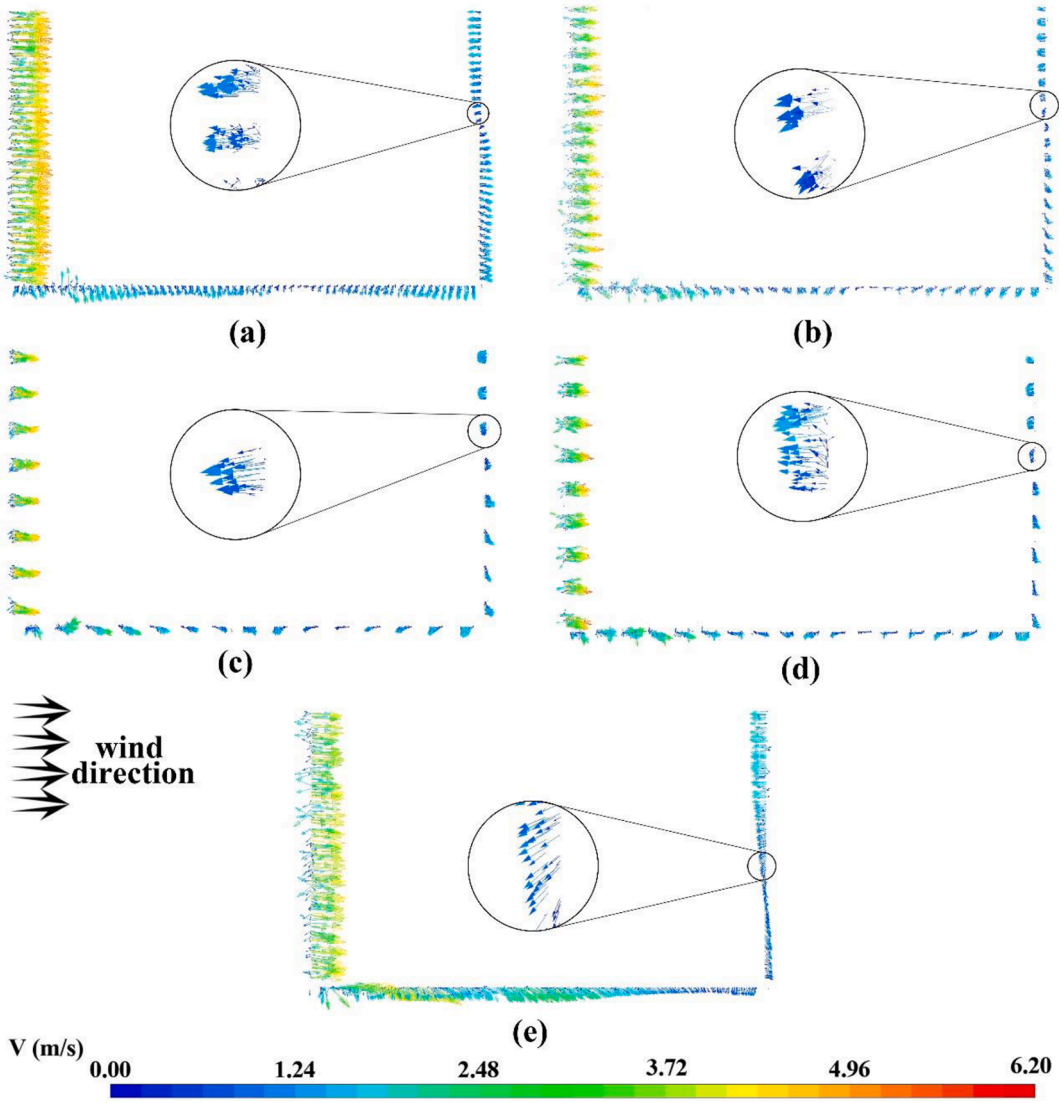


Fig. 13. Velocity vector in three sides of frame in (a) frame 3 (b) frame 4 (c) frame 5 (d) frame 6 (e) frame 7, at V = 4 m/s.

e" are not desirable as a large part of the frame surface heat transfer area is removed.

- In natural convection, perforations can noticeably reduce the average module temperature. By employing frame 4, a 3.8 K reduction is observed with $q''_{eq} = 600$ W/m^2, in comparison to the conventional frame 1.
- With the typical ambient conditions in a year, frame 4 achieves about 0.6% more power in case of natural convection.
- Finally, frame 4 is proposed as an alternative to the conventional frame 1. It does not cause any noticeable temperature increase in windy ambient, while it cools the module in nearly zero wind speed. Additionally, aluminum consumption is reduced by 7%.

Although the resulting improvements seem small, it can lead to a large amount of extra electricity generation and aluminum conservation, if the proposed perforations are applied on large-scale PV power plants. Further studies in different weather conditions are necessary to gain a better perspective on the economical and also mechanical aspects of modified large-scale PV power plants.

CRediT authorship contribution statement

S. Nazari: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Visualization.

M. Eslami: Conceptualization, Methodology, Resources, Writing - original draft, Writing - review & editing, Supervision, Project administration.

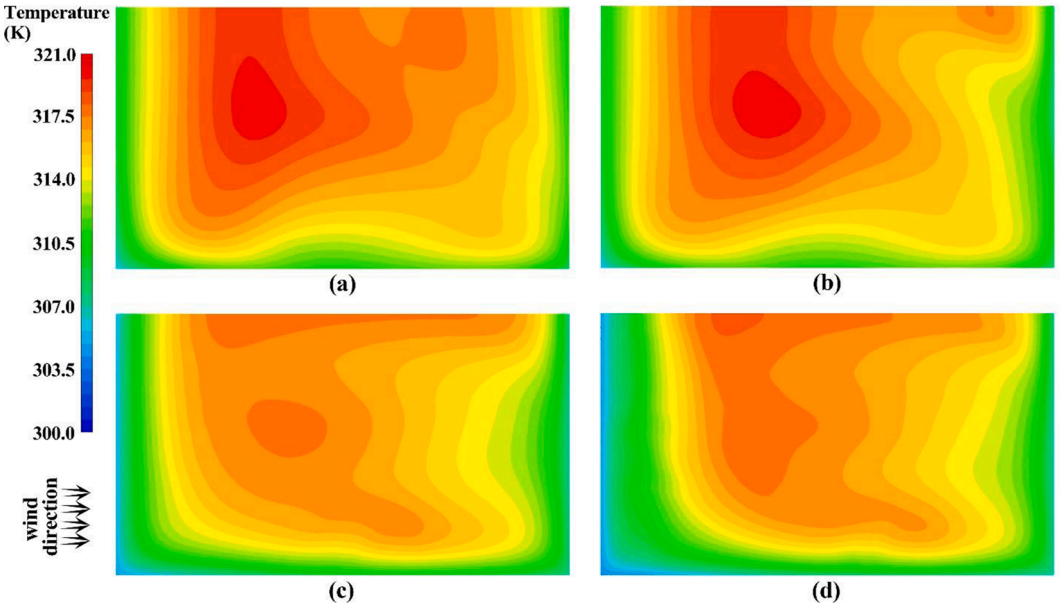


Fig. 14. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 8 (c) frame 4 (d) frame 9, at V = 4 m/s.

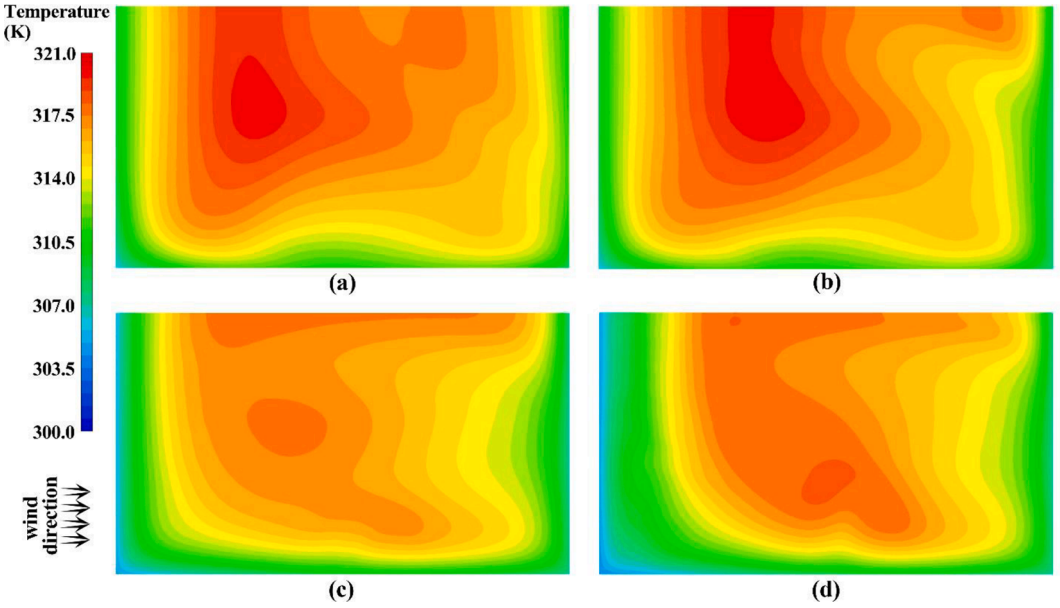


Fig. 15. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 10 (c) frame 4 (d) frame 11, at V = 4 m/s.

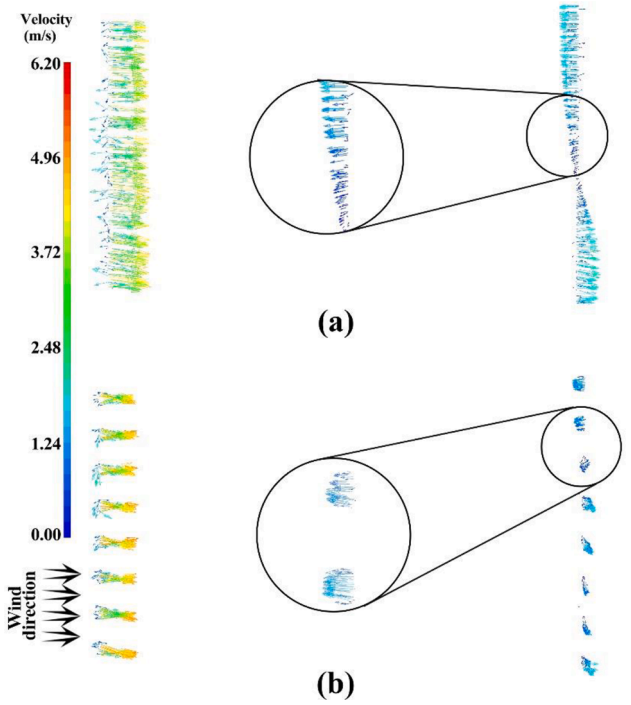


Fig. 16. Velocity vector in (a) frame 12 (b) frame 13, at V = 4 m/s.

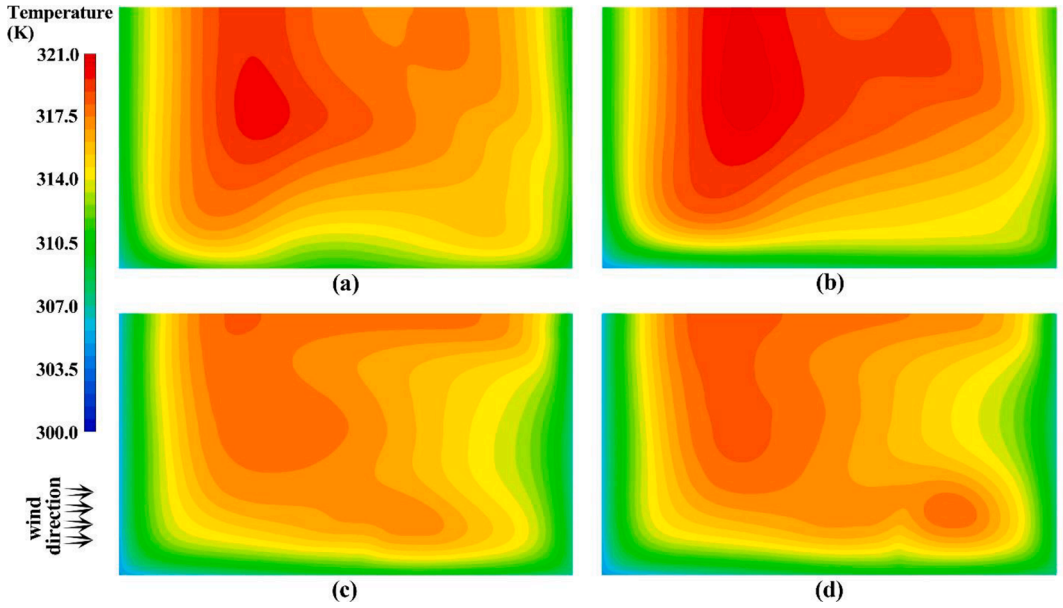


Fig. 17. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 12 (c) frame 5 (d) frame 13, at V = 4 m/s.

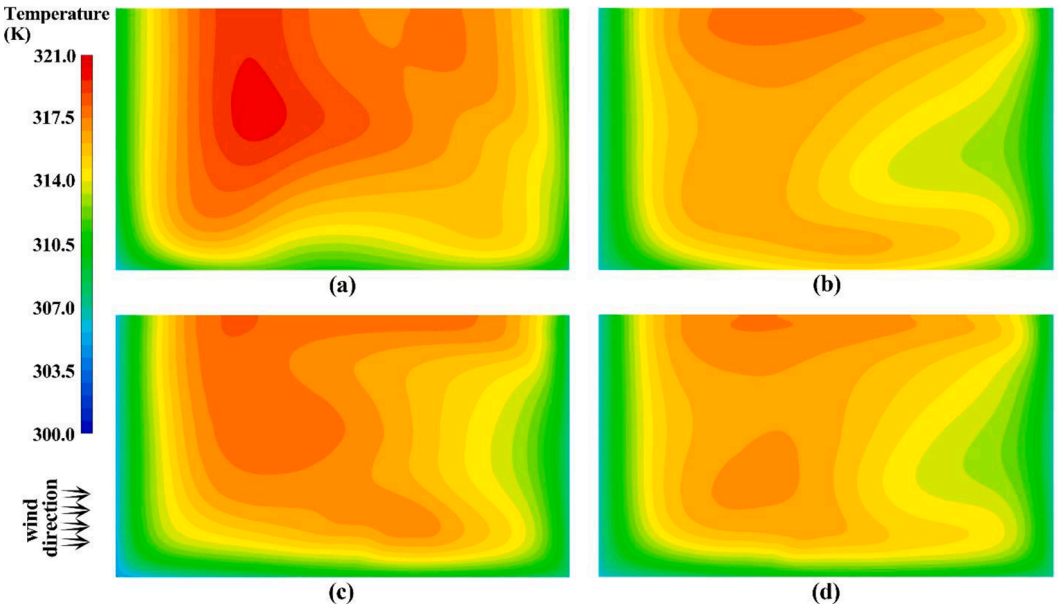


Fig. 18. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 14 (c) frame 5 (d) frame 15, at V = 4 m/s.

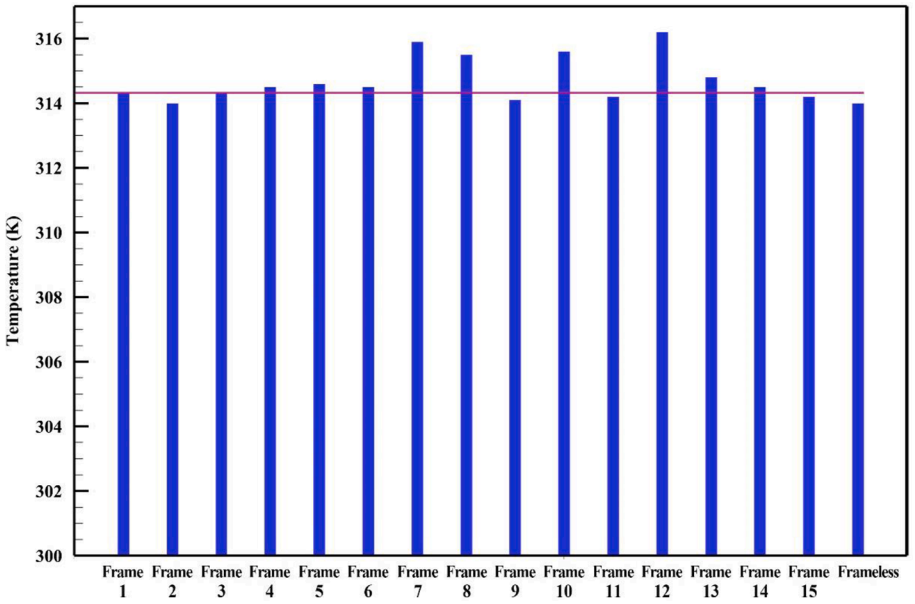


Fig. 19. Averaged PV temperature in proposed frame designs at V = 4 m/s.

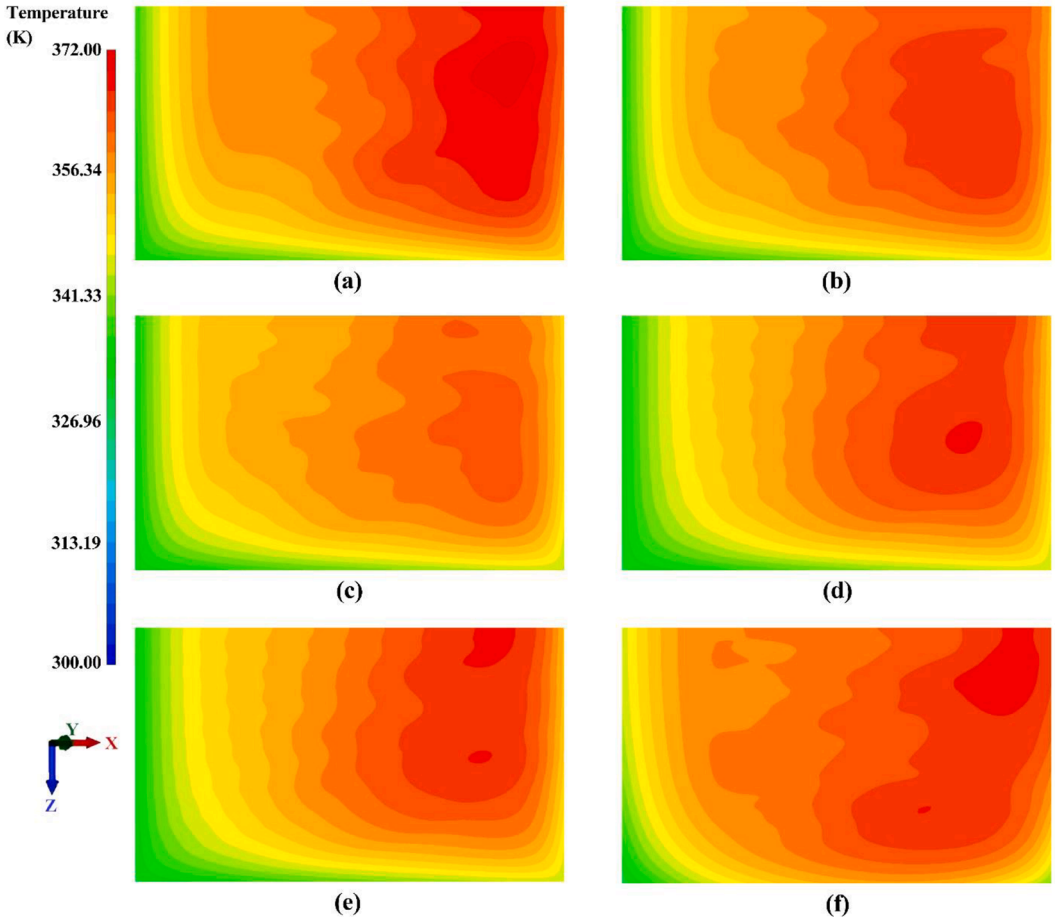


Fig. 20. Temperature distribution on back surface of PV module with (a) frame 1 (b) frame 3 (c) frame 4 (d) frame 5 (d) frame 6 (d) frame 7, at V = 0 m/s.

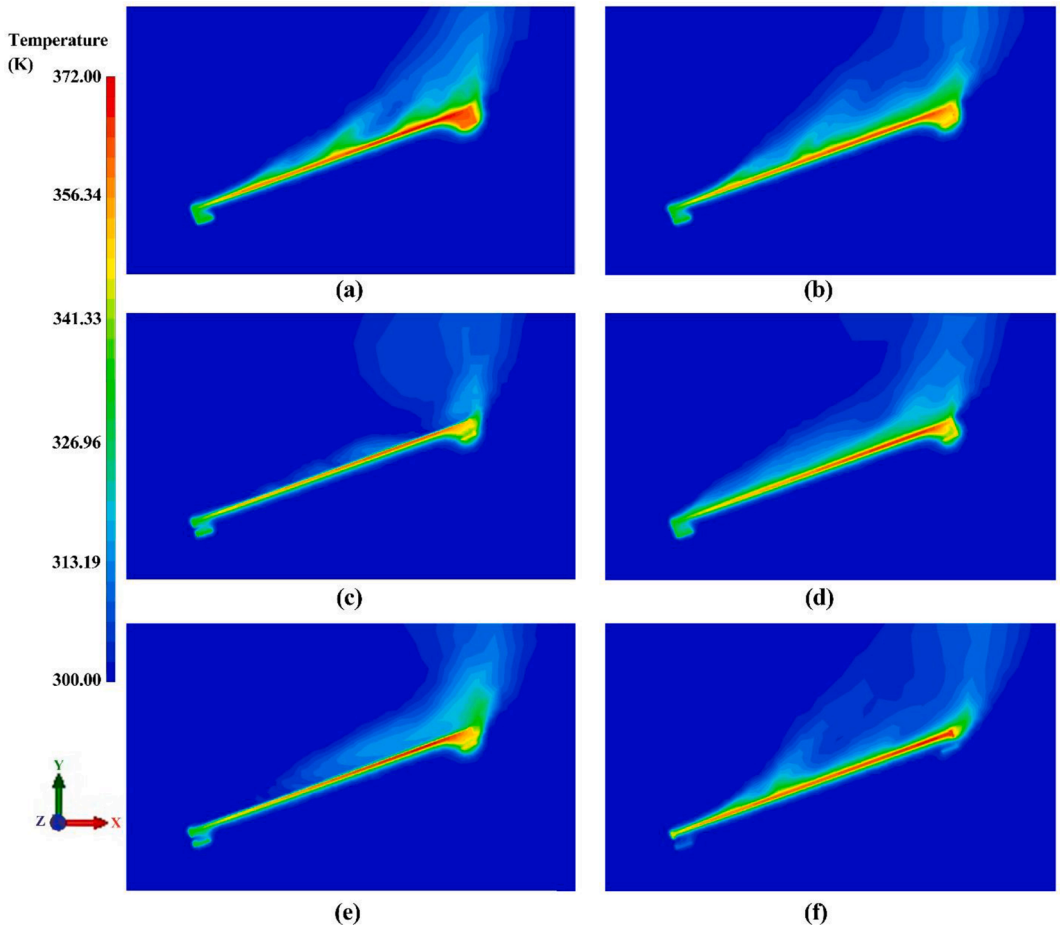


Fig. 21. Temperature distribution on symmetry surface of PV module with (a) frame 1 (b) frame 3 (c) frame 4 (d) frame 5 (e) frame 6 (f) frame 7, at V = 0 m/s.

Table 9
Heat transfer results of perforated frames at V = 0 m/s.

Perforated side	Design name	T_{avg} (K)	h_{back} (W/m^2.K)	h_{front} (W/m^2.K)	Q_{frame} (W)	h_{frame} (W/m^2.K)	A_{frame} (cm^2)
Three sides	Frame 1	358.6	2.9	5.2	24.4	3.6	1648.5
	Frame 3	357.2	2.9	5.3	23.8	3.9	1657.9
	Frame 4	354.8	2.9	5.7	25.4	4.3	1591.4
	Frame 5	355.5	2.9	5.5	25.5	4.4	1573.1
	Frame 6	355.7	2.9	5.5	25.1	4.2	1585.4
	Frame 7	360.9	3.2	5.5	14.8	5.2	1383.1
	Frame 8	359.2	3.1	5.4	20.3	2.6	1428.4
	Frame 9	356.1	2.9	5.6	23.4	3.9	1600.5
Two sides	Frame 10	360.4	3.0	5.3	19.9	4.0	1455.2
	Frame 11	356.8	2.9	5.5	23.4	3.7	1608.2
	Frame 12	358.3	3.1	5.2	22.5	4.8	1504.3
	Frame 13	355.0	2.9	5.6	25.0	4.3	1609.1
One side	Frame 14	360.3	3.0	5.2	21.6	3.7	1527.3
	Frame 15	356.7	2.9	5.5	24.0	3.7	1612.6

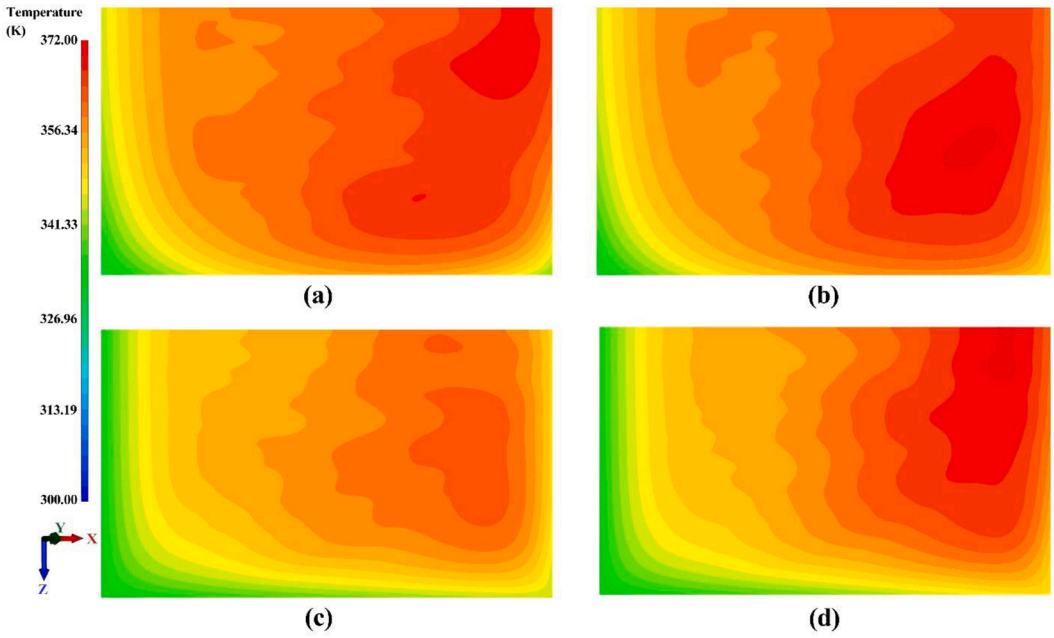


Fig. 22. Temperature distribution on backside plane of PV module with (a) frame 7 (b) frame 10 (c) frame 4 (d) frame 11, at V = 0 m/s.

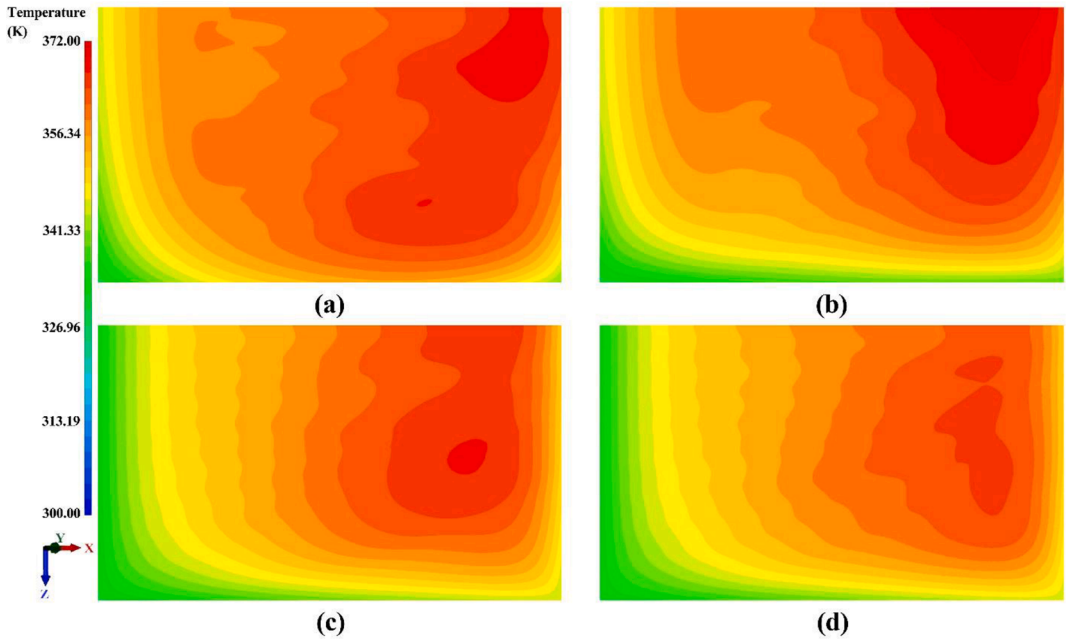


Fig. 23. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 12 (c) frame 5 (d) frame 13, at V = 0 m/s.

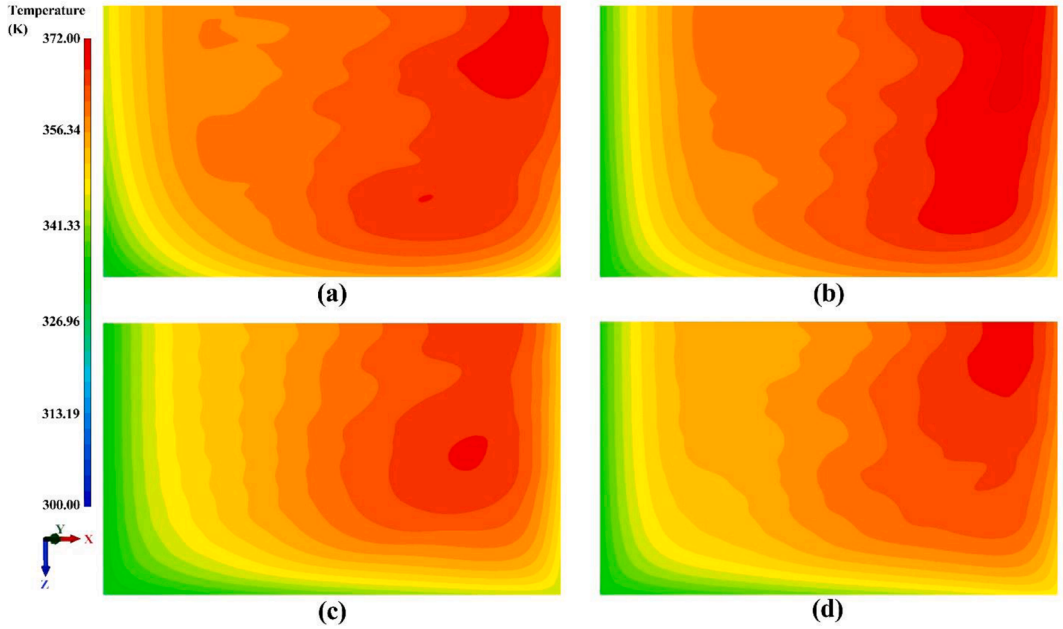


Fig. 24. Temperature distribution on back surface of PV module with (a) frame 7 (b) frame 14 (c) frame 5 (d) frame 15, at V = 0 m/s.

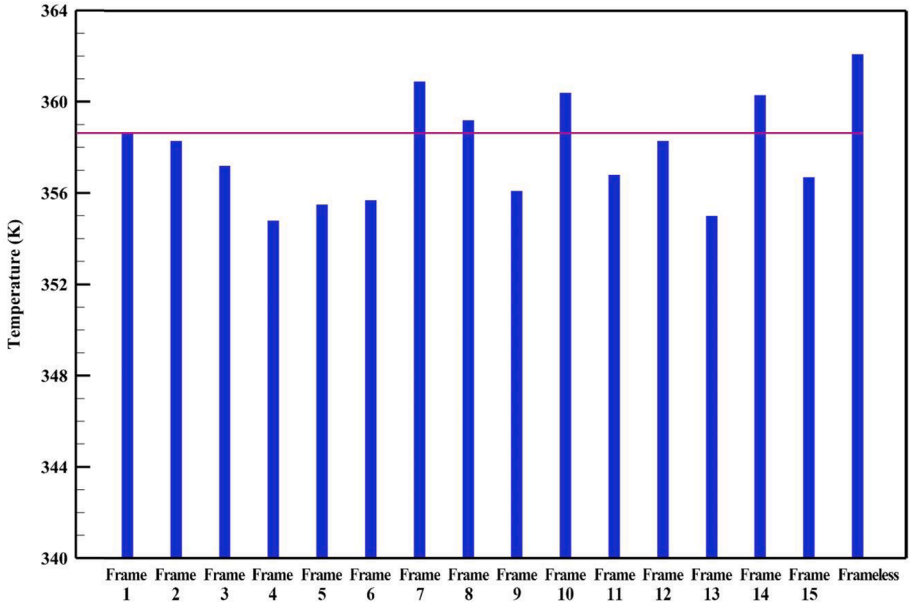


Fig. 25. Averaged PV temperature in proposed frame designs at V = 0 m/s.

Table 10
Typical weather conditions for three days in Shiraz, Iran [26].

Day	Total incident solar irradiance (W/m^2)	Ambient temperature (K)
Winter	580	270.6
Temperate	970	283.6
Summer	1057	312

Table 11
Performance of selected frame designs in typical weather conditions.

Design	V = 4 m/s			V = 0 m/s		
	T_{avg} (K)	$\frac{P_{out}}{A}$(W/m^2)	q''_{eq}(W/m^2)	T_{avg} (K)	$\frac{P_{out}}{A}$(W/m^2)	q''_{eq}(W/m^2)
Winter weather condition						
Frame 1	278.5	87.8	332.5	297	80.8	219
Frame 4	278.5	87.8	331.5	296	81.2	225.5
Frame 9	278.4	87.9	333	296.2	81.1	224
Temperate weather condition						
Frame 1	299.2	148.2	660	327.7	128.2	440
Frame 4	299.3	148.2	658.3	326.6	129.0	449.85
Frame 9	299.1	148.3	661	327.3	128.4	442.75
Summer weather condition						
Frame 1	326.2	116.6	601.2	350.6	101.2	358
Frame 4	326.3	116.6	599	349.6	101.8	369.25
Frame 9	326.1	116.7	602	350.1	101.6	364.25

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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